



# Poisson Distribution · Mark Scheme

Further Statistics 1 Chapter 2: Poisson Distribution. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

## Section A – Poisson probabilities and conditions

- 1  $P(X = 2) = \frac{e^{-3}3^2}{2!}$  M1; = 0.2240 A1.
- 2 (a)  $P(X \leq 3) = 0.3423$  B1; (b)  $1 - P(X \leq 5)$  M1; =  $1 - 0.7029 = 0.2971$  A1.
- 3 Any three, in context: emails arrive singly (one at a time) B1; emails arrive independently of each other B1; emails arrive at random, at a constant average rate B1.
- 4 (a)  $e^{-\lambda} = 0.2$  M1;  $\lambda = \ln 5$  A1. (b)  $P(X = 2) = 0.2 \times \frac{(\ln 5)^2}{2}$  M1; = 0.2590 A1.
- 5(a) Customers arrive singly and independently B1; at a constant average rate B1.
- 5(b)  $\frac{e^{-6}6^4}{4!}$  M1; = 0.1339 A1.
- 5(c)  $1 - P(X \leq 8)$  M1; =  $1 - 0.8472 = 0.1528$  A1.
- 5(d) For a Poisson distribution the mean equals the variance B1; here the variance (11.2) is much larger than the mean (6.1), so the model is not suitable B1.

## Section B – Adding and changing the interval

- 6  $Po(0.4 \times 5) = Po(2)$  M1;  $P(0) = e^{-2} = 0.1353$  A1.
- 7  $X + Y \sim Po(5)$  B1;  $\frac{e^{-5}5^4}{4!}$  M1; = 0.1755 A1.
- 8  $Po(18)$  B1;  $1 - P(X \leq 20)$  M1; =  $1 - 0.7307 = 0.2693$  A1.
- 9 (a)  $T \sim Po(4)$  M1;  $1 - P(T \leq 2) = 1 - 0.2381 = 0.7619$  A1. (b)  $Po(8)$  M1;  $\frac{e^{-8}8^5}{5!} = 0.0916$  A1.
- 10(a) Total errors per page  $E \sim Po(1)$  M1;  $P(E = 0) = e^{-1} = 0.3679$  A1.
- 10(b)  $Po(8)$  B1;  $1 - P(X \leq 10)$  M1; =  $1 - 0.8159 = 0.1841$  A1.
- 10(c)  $\frac{P(T = 2)P(S = 0)}{P(E = 2)}$  M1; =  $\frac{e^{-0.6} \frac{0.6^2}{2} \times e^{-0.4}}{e^{-1} \frac{1^2}{2}}$  A1; = 0.36 A1.
- 10(d)  $Y \sim B(5, 0.3679)$  M1;  $1 - (0.6321)^5 - 5(0.3679)(0.6321)^4 = 0.6054$  A1.

## Section C – Poisson approximation

- 11  $n$  large (e.g.  $n > 50$ ) and  $p$  small (e.g.  $p < 0.1$ ) B1;  $X \approx Po(np)$  B1.
- 12  $\lambda = 200 \times 0.015 = 3$  B1;  $\frac{e^{-3}3^2}{2!}$  M1; = 0.2240 A1.
- 13  $Po(3)$  B1;  $1 - P(X \leq 4)$  M1; =  $1 - 0.8153 = 0.1847$  A1.
- 14 (a)  $0.95^{60} + 60(0.05)(0.95)^{59} + \binom{60}{2}(0.05)^2(0.95)^{58}$  M1; = 0.4174 A1. (b)  $Po(3)$  M1;  $P(X \leq 2) = 0.4232$  A1.
- 15(a)  $Po(3)$  B1, since  $n = 200$  is large and  $p = 0.015$  is small B1.
- 15(b)  $\frac{e^{-3}3^4}{4!}$  M1; = 0.1680 A1.
- 15(c)  $P(X > 5) = 1 - P(X \leq 5)$  M1; =  $1 - 0.9161$  A1; = 0.0839 A1.
- 15(d)  $R \sim B(10, 0.0839)$  M1;  $(0.9161)^{10} + 10(0.0839)(0.9161)^9$  M1; = 0.7975 A1.



## Roar-level challenge

16 (a)  $\frac{P(X=2)P(Y=3)}{P(X+Y=5)}$  with  $X+Y \sim \text{Po}(5)$   $M1; = \frac{e^{-2} \frac{2^2}{2!} \times e^{-3} \frac{3^3}{3!}}{e^{-5} \frac{5^5}{5!}}$   $M1; = 0.3456$   $A1$ . (b)  $\frac{e^{-\lambda} \frac{\lambda^r}{r!} e^{-\mu} \frac{\mu^{n-r}}{(n-r)!}}{e^{-(\lambda+\mu)} \frac{(\lambda+\mu)^n}{n!}}$   $M1; =$

$\binom{n}{r} \left( \frac{\lambda}{\lambda+\mu} \right)^r \left( \frac{\mu}{\lambda+\mu} \right)^{n-r}$ , the  $B\left(n, \frac{\lambda}{\lambda+\mu}\right)$  probability function  $A1$ .